

CARLOS PEREZ MENDOZA

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Professional summary

AI Scientist with a Ph.D. in Mathematics & Statistics and 8+ years of experience applying machine learning, reinforcement learning, and probabilistic modeling to quantitative finance, risk management, and healthcare. Proven ability to design and deploy scalable, high-performance AI systems that deliver measurable impact in algorithmic trading, financial risk analytics, and digital health innovation.

SKILLS & EXPERTISE

Domains of Application: Finance, Risk Management, Digital Health.

- Deep Learning (PyTorch, TensorFlow)
- Probabilistic Modelling
- Data Management (Python, SQL, R)
- Reinforcement Learning
- Statistical Modelling
- Data Visualization
- Bayesian Optimization
- Quantitative Analysis
- Signal Processing

SELECT ACHIEVEMENTS

Published in top peer-reviewed journals across quantitative finance, actuarial science, and digital health, applying deep learning, reinforcement learning, and AI models to solve high-impact problems in these domains.

Professional Experience

Postdoctoral Fellow / AI Research & Applied Data Science

McGill University

Montreal, Canada

Since Oct 2024

- Design and implement end-to-end ML pipelines that achieve ~80% accuracy in cardiovascular disease screening and biomarker detection across multiple clinical trials, delivering actionable scientific reports to pharmaceutical partners, digital health companies, and other stakeholders.

Scientifique de recherche appliquée

JACOB - Centre d'intelligence artificielle appliquée, Direction Recherche & Innovation

Montreal, Canada

Mar 2023-Jul 2024

- Designed and implemented AI-driven pipelines for startup-investor matchmaking, startup recommendation, and wearable-based diabetes monitoring, leveraging NLP, web scraping, and deep learning (transformers), achieving accuracy rates of 83%, 85% and 87%, respectively.

Quantitative Research Analyst

Plant-E Corp, Algorithmic Trading Department

Montreal, Canada

Jul 2022-Nov 2022

- Developed and implemented machine learning-based methodologies for quantitative trading and electricity price forecasting in North American Power Markets, leveraging reinforcement learning to improve day-ahead and futures price forecasting precision by 5% while meeting regulatory requirements.

Presidential Advisor

Private Assistance Board of the Government of Mexico City

Mexico City

Oct 2020-Ago 2021

- Led ML projects to identify financial risk across 100+ Private Assistance Institutions in Mexico City, strengthening long-term sustainability. Implemented data-driven systems that enhanced board-level decision-making and reduced inefficiencies by 20%.

Head of the Office of Quantitative Health Plan Studies

Bank of Mexico (Central Bank of Mexico)

Mexico City

Nov 2017-Jan 2019

- Coordinated and designed actuarial models and automated data systems for the Bank of Mexico's medical service, optimizing resource allocation and improving operational accuracy by 30%, while strengthening long-term sustainability.

Market Research Analyst

Bank of Mexico (Central Bank of Mexico)

Mexico City

Mar 2015-Oct 2017

- Implemented anomaly detection and predictive models for the Bank of Mexico's medical service, preventing fraud with monthly recoveries of 2.5% of drug turnover, reducing annual health expenditures by 1.23%, and optimizing supplier performance to reach 98% drug coverage and 2.3% annual savings.

Education

Ph.D. Mathematics & Statistics | Concordia University, Canada | Sep 2021–Apr 2025

M.Sc. Mathematics, Mathematical Statistics & Probability | UNAM, Mexico | Jan 2019–Jul 2020

M.Sc. Modern Applications of Mathematics | University of Bath, UK | Jan 2020–Jul 2020

Specialist Degree in Financial Engineering | UNAM, Mexico | Aug 2021–Jul 2022

B.Sc. Actuarial Sciences – UNAM, Mexico | Aug 2010–Dec 2014

Graduate Diploma in Statistical Techniques & Data Mining | UNAM, Mexico | Jan 2016–Oct 2016

Certification & Training

Artificial Intelligence: An Overview Specialization (5 Courses) – Politecnico di Milano, 2023

Introduction to Data Science in Python – University of Michigan, 2020

Introduction to TensorFlow for AI, ML, and DL – DeepLearning.AI, 2019

Deep Learning Specialization (5 Courses) – DeepLearning.AI, 2018

Quantum Computing 101 – IBM, 2018

Microsoft Excel and Visual Basic for Applications (VBA)

Languages

English: Full Professional

French: Intermediate

Spanish: Native